AI-Powered Medical Report Interpretation System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstraction

The rapid increase in clinical documentation poses a significant challenge for healthcare professionals, often leading to information overload and diagnostic delays. Traditional Hospital Information Systems (HIS) function as static databases, requiring doctors to manually scan through voluminous reports to identify critical abnormalities. This research proposes an **AI-Powered Medical Report Interpretation System**, specifically focusing on a **Doctor-Oriented Clinical AI Chatbot** designed to provide secure, evidence-based decision support.

The system utilizes a **Retrieval-Augmented Generation (RAG)** framework, ensuring that the AI's responses are strictly grounded in verified patient medical reports, thereby mitigating the risk of "hallucinations" common in generic Large Language Models (LLMs). A core focus of this research is the integration of a **privacy-first architecture**, incorporating automated **De-identification** of Personally Identifiable Information (PII) using Named Entity Recognition (NER) and **Role-Based Access Control (RBAC)** to ensure data security and HIPAA/GDPR compliance.

The chatbot is further optimized with a domain-specific **Natural Language Processing (NLP)** engine, capable of interpreting complex medical shorthand (e.g., CBC, BUN, CrCl) and maintaining a professional "doctor-to-doctor" communication tone. Evaluation of the system focuses on performance metrics such as hallucination rates, response latency, and efficiency gains in diagnostic speed. By bridging the gap between raw laboratory data and actionable insights, this research contributes to UN Sustainable Development Goal 3 (Good Health and Well-being) by enhancing clinical workflow efficiency and patient safety through explainable and secure AI.

Keywords: *Retrieval-Augmented Generation (RAG), Clinical AI Chatbot, Medical Report Interpretation, De-identification, Role-Based Access Control (RBAC), Healthcare Informatics*.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
AD	Active Directory (Azure)
API	Application Programming Interface
BUN	Blood Urea Nitrogen
CBC	Complete Blood Count
CNLP	Clinical Natural Language Processing
CrCl	Creatinine Clearance
DSR	Design Science Research
EHR	Electronic Health Record
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
HIS	Hospital Information System
IEEE	Institute of Electrical and Electronics Engineers
LFT	Liver Function Test
LLM	Large Language Model
NER	Named Entity Recognition
NIC	National Identity Card
NLP	Natural Language Processing
PII	Personally Identifiable Information
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
SDG	Sustainable Development Goal
SSL	Secure Sockets Layer
TLS	Transport Layer Security
TLS	User Interface
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Background and Context

Modern healthcare systems face a significant challenge in managing the sheer volume of clinical documentation. Medical practitioners are frequently overwhelmed by dense laboratory reports and complex clinical data, which require rapid and precise interpretation. The manual processing of these reports is not only labor-intensive but also susceptible to cognitive fatigue, which can result in "information overload" and increased diagnostic latency [1].

This issue primarily affects healthcare professionals (doctors, consultants, and interns) and patients. For clinicians, the time spent on manual data synthesis contributes to professional burnout and reduces the time available for direct patient interaction. For patients, any delay or error in report interpretation can lead to a postponement of critical medical interventions, directly impacting the quality of care and safety [2].

Recent global estimates in 2024 suggest that diagnostic errors contribute to approximately 2.59 million missed diagnoses annually, resulting in nearly 795,000 cases of serious harm [3]. Furthermore, studies indicate that nearly 16% of preventable harm in healthcare systems worldwide is linked to delayed or incorrect diagnoses [4]. In regional contexts like Sri Lanka, where the patient-to-doctor ratio is high, the administrative burden on doctors is even more acute, making automated assistance a necessity rather than a luxury.

Current solutions generally fall into few categories:

1. **Generic LLMs:** Tools like ChatGPT are often used but lack clinical-grade safety, frequently exhibiting "hallucinations" where they generate plausible but medically incorrect information [5].
2. **Patient-Facing Chatbots:** These are designed for simple symptom checking or emotional support and do not support the professional medical terminology or structured analysis required by a doctor [5].
3. **Traditional EHR Systems:** Most existing HIS serve as static databases. They lack the interactive capability to allow a doctor to query specific abnormal findings across multiple reports dynamically.

To overcome the limitations of the mentioned systems, researchers have transitioned from traditional **Supervised Fine-Tuning (SFT)** to **Retrieval-Augmented Generation (RAG)**. Traditional model training involves retraining a Large Language Model (LLM) on a specific medical dataset, which is computationally expensive and leads to "knowledge cutoff" where the model cannot access real-time patient data [19]. Furthermore, fine-tuned models often suffer from "black-box" reasoning, making it difficult for clinicians to trace the source of a specific medical suggestion [20].

In contrast, the RAG-based approach decouples the LLM's reasoning capabilities from its internal memory. By retrieving context directly from verified clinical documents and feeding it to the generator (e.g., GPT-4o), the system ensures that responses are grounded in "ground-truth" data rather than stochastic predictions [21]. This methodology significantly reduces the risk of medical hallucinations a critical requirement for clinical-grade AI and allows for seamless updates to patient records without the need for constant model retraining [19], [22].

The proposed AI-Powered Medical Report Interpretation System is a secure, multi-modular platform designed to bridge the gap between complex clinical data and rapid medical decision-making. While the system operates as an integrated ecosystem, my research focuses specifically on the development and optimization of the **Doctor-Oriented Clinical AI Chatbot**.

The following components define the technical and ethical framework of the solution:

- **Secure Doctor Qualification & Access Control(Member 1) :** Provides the fundamental security layer by utilizing Azure AD for institutional verification and medical license validation.
- **Clinical Record Management (Member 2):** Ensures data integrity through auditable change tracking, feeding the chatbot with verified and synchronized patient histories.
- **Doctor-Oriented Clinical AI Chatbot (Member 3):** This is a specialized conversational interface designed strictly for professional medical use. The chatbot is distinguished by the following core technical strategies:
    - **Retrieval-Augmented Generation (RAG):** Unlike traditional "black-box" AI, this chatbot generates responses based solely on the verified medical reports uploaded to the system. This eliminates hallucinations and ensures that every summary is evidence-based and traceable to a specific source document.
    - **Clinical Linguistic Alignment:** The chatbot is fine-tuned to understand and utilize complex medical terminology and standard clinical abbreviations (e.g.,

CBC, BUN, CrCl). It maintains a professional, neutral, and data-driven tone consistent with doctor-to-doctor clinical handovers.

    - **Data Privacy & De-identification:** To ensure patient confidentiality, the chatbot engine employs a de-identification strategy, where sensitive Personal Identifiable Information (PII) is masked during the processing phase.
    - **Advanced Security Protocols:** The interface utilizes Encrypted Communication (TLS/SSL) for all data in transit and maintains rigorous Audit Logging of every query made by the doctor to ensure accountability.
    - **Patient-Doctor Role-Based Access (RBAC):** The chatbot's data retrieval is strictly governed by the doctor's authorization level. A doctor can only query and retrieve insights for patients directly under their care, preventing unauthorized cross-access to sensitive medical records.

- **AI-Assisted Report Analysis Engine (Member 4) :** Acts as the data processing backbone that extracts numeric values and flags abnormalities, which the chatbot then interprets and presents to the clinician.

This research directly aligns with United Nations Sustainable Development Goal 3: Good Health and Well-being, specifically Target 3.8 (Achieving universal health coverage) and Target 3.c (Increasing health financing and support for the health workforce in developing countries) by enhancing clinical efficiency through digital health technologies [4].

The project is situated within the Health Informatics and Applied Artificial Intelligence research clusters, focusing on Explainable AI (XAI) and secure clinical decision support systems.

1.2 Literature Review

Recent advancements in Generative AI have shifted from general-purpose models to domain-specific clinical assistants. Research in 2024 emphasizes that while Large Language Models (LLMs) like GPT-4 show high competency in medical exams, they lack the "grounding" required for patient-specific reasoning [6]. To solve this, the **Retrieval-Augmented Generation (RAG)** framework has emerged as a gold standard in 2025 for clinical applications. RAG allows the system to retrieve data from verified, uploaded medical reports before generating a response, which significantly reduces the risk of "hallucinations" [7].

Furthermore, scholarly work on Clinical Natural Language Processing (CNLP) highlights the importance of "Medical Terminology Mapping." Recent studies have focused on fine-tuning models to recognize complex abbreviations (e.g., *TAVR, CBC, CrCl*) and maintaining a

professional "Doctor-to-Doctor" tone, which differs significantly from the empathetic but simplified tone used in patient-facing bots [8], [9]. Security-centric research also underlines the necessity of **De-identification** and **Role-Based Access Control (RBAC)** to ensure that clinical chatbots comply with international health data regulations like HIPAA and GDPR [10].

Several products have entered the healthcare AI market recently:

- **General Purpose LLMs (ChatGPT, Gemini):** Frequently used by doctors for drafting letters or summarizing notes, but they lack institutional security and can produce non-existent clinical citations [11].
- **Microsoft Dragon Ambient eXperience (DAX) / Copilot:** Focuses heavily on "notetaking" and "medical scribing" during consultations rather than interactive report analysis [12].
- **ChatRWD:** A specialized RAG system that has shown up to a **58% improvement** in answering evidence-based clinical questions compared to standard LLMs [13].
- **Patient-Facing Bots (e.g., Ada, Babylon):** These are optimized for symptom checking and triage using simple language but are not designed for the high-level technical analysis required by a consultant [14].

Despite these advancements, existing systems exhibit several critical gaps that this research aims to address:

1. **Hallucination and Unreliability:** Standard LLMs often create "plausible-sounding" but medically incorrect values if they are not strictly grounded in a patient's specific report [6], [11].
2. **Privacy and Anonymization Gaps:** Most commercial chatbots do not automatically de-identify PII (Personally Identifiable Information) before processing, posing a massive data breach risk in clinical settings [10].
3. **Lack of Specialized Tone:** Existing interfaces are either too "robotic" or too "patient-friendly," failing to match the precise, abbreviation-heavy language doctors use for efficient communication [8].
4. **Static Data Retrieval:** Traditional Hospital Information Systems (HIS) do not allow for conversational queries, forcing doctors to manually hunt for data across pages of reports [15].

1.3 Research Gap

Current AI implementations in healthcare primarily focus on administrative automation or patient-facing symptom checkers. There is a distinct lack of a **dedicated conversational interface designed exclusively for clinical doctors** that can interact with patient-specific laboratory data in real-time. Most existing systems provide static summaries but fail to offer a dynamic "Q&A" environment where a doctor can query specific abnormal trends across multiple reports using professional medical shorthand [1], [15].

Existing approaches are insufficient due to several critical factors identified in recent research:

- **Contextual Blindness & Hallucinations:** Traditional Large Language Models (LLMs) operate on general knowledge and often generate "hallucinations" medically incorrect data when applied to specific patient reports [10].
- **Privacy & Security Gaps:** Current commercial chatbots do not implement automated **de-identification (masking PII)** or institutional **Role-Based Access Control (RBAC)** at a granular level, making them non-compliant with clinical data governance standards [10], [16].
- **Linguistic Mismatch:** Most AI models are tuned for a general or empathetic tone and lack the specialized capability to interpret complex medical abbreviations and the high-density language used in professional doctor-to-doctor handovers [2], [5].
- **Static Data vs. Real-time Interaction Gap:** Most existing Hospital Information Systems (HIS) and Electronic Health Records (EHR) function as static databases. While they store vast amounts of patient data, they lack a conversational interface that allows a doctor to query specific information dynamically. Currently, clinicians must manually navigate through multiple pages or tabs to find specific lab values, creating a significant "interaction bottleneck" that delays clinical decision-making [15], [18].
- **Static Training vs. Dynamic Data Interpretation:** Traditional AI models often rely on a "static knowledge base" created during their initial training phase. In clinical settings, where patient reports are generated in real-time, these models cannot incorporate new or updated laboratory findings without undergoing extensive and costly retraining. This research addresses this by utilizing the RAG framework, which allows for immediate, "source-grounded" interpretation of any new document, ensuring the AI's reasoning is always based on the most current data rather than outdated training sets [6], [7].

Table 1. Comparative Analysis of Clinical AI Architectures

Feature	Static EHR / HIS Systems	Generic LLMs (e.g., ChatGPT)	Supervised Fine-Tuning (SFT)	Proposed RAG System	Supporting References
Real-time Data Access	✓	✗	✗	✓	[7], [21]
Medical Fact Accuracy	✓	✗	Partial	✓	[13], [20]
Source Traceability	✗	✗	✗	✓	[19], [22]
PII Data Masking	✗	✗	✗	✓	[10], [16]
Computational Cost	Low	Low	High	Medium	[19], [21]
Knowledge Update Speed	Instant	✗	Very Slow	Instant	[7], [19]
Clinical Tone (Shorthand)	✗	Partial	✓	✓	[8], [9]
Granular Security (RBAC)	✓	✗	✗	✓	[16], [17]

1.4 Novelty of the Research

The novelty of this component lies in the integration of a specialized **Retrieval-Augmented Generation (RAG)** framework designed exclusively for doctor-oriented clinical decision support. Unlike general-purpose AI models that operate on broad, unverified data, this system is strictly grounded in verified hospital documentation. This ensuring that every generated response is technically accurate, traceable to a specific source, and evidence-based [7].

The creativity and uniqueness of the solution are further demonstrated through the following technical implementations:

- **Clinical Linguistic Alignment:** Beyond simple conversation, the chatbot is fine-tuned to interpret complex medical shorthand (e.g., CBC, CrCl, BUN) and maintain a professional "doctor-to-doctor" tone, which is a significant departure from standard empathetic or simplified AI interfaces.
- **Privacy-First Conversational Engine:** By combining high-level encrypted communication and audit logging with an automated de-identification layer, the system ensures that sensitive Personal Identifiable Information (PII) is masked. This

ensures that clinical data is never exposed to the underlying Large Language Model (LLM) in a readable format.

- **Granular Security Integration:** The system enforces a strict **Role-Based Access Control (RBAC)** mechanism at the query level. This restricts a doctor's access only to the records of patients specifically assigned to them, addressing a critical security gap and representing a sophisticated level of data governance rarely seen in undergraduate-level AI research [16], [17].
- **Dynamic Evidence Attribution:** Every summary provided by the chatbot is linked with automated citations to the source document, providing the "Explainable AI" (XAI) transparency required for high-stakes clinical decision-making.

1.5 Justification for the Proposed Solution (Evidence-based Argument)

The selection of a RAG-based architecture for this research is supported by current clinical informatics trends and empirical evidence:

- **Addressing Clinician Trust**: Research confirms that while AI accuracy is improving, a significant percentage of clinicians remain hesitant to use AI due to concerns regarding data privacy and the potential for misinformation [10]. This project directly addresses this barrier by implementing localized de-identification and private data processing.
- **Mitigating Hallucinations**: Evidence suggests that a specialized, grounded Retrieval-Augmented Generation (RAG) framework can improve clinical decision-making speed while maintaining a near-zero hallucination rate when compared to generic models [13]. This justifies the use of RAG as a superior alternative to standard LLMs for high-stakes medical decision support.
- **Dynamic Knowledge vs. Static Training:** Unlike traditional models that rely on a static knowledge base, the RAG framework allows for the interpretation of real-time clinical reports. Since medical data is generated continuously, RAG ensures that the AI's reasoning is based on the most current laboratory findings rather than outdated training sets [7], [21].
- **Explainability and Traceability (Source Grounding):** A key challenge in clinical AI is the "black-box" nature of large language models. RAG addresses this by providing explicit citations for every summary generated, allowing clinicians to trace AI insights back to the specific page and line of the source medical report, thereby enhancing diagnostic accountability [19], [22].
- **Handling Hallucination in High-Stakes Environments:** Generic LLMs are prone to hallucinations when faced with complex medical jargon not present in their training

data [20]. By utilizing a "retrieval-first" approach, this system constrains the AI's response to the retrieved context, significantly reducing the risk of generating medically inaccurate information [13], [20].

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a secure, **Retrieval-Augmented Generation (RAG) based clinical conversational assistant** that enables qualified doctors to interactively query and summarize patient-specific medical reports with a 95% reduction in information hallucinations compared to general-purpose LLMs.

2.2 Specific Objectives

- **To implement a Retrieval-Augmented Generation (RAG) pipeline** that strictly grounds the chatbot's responses in verified medical reports, ensuring every clinical summary is traceable to a specific source document.
- **To develop a domain-specific Natural Language Processing (NLP) engine** capable of accurately interpreting complex medical abbreviations (e.g., CBC, BUN, CrCl) and maintaining a professional "doctor-to-doctor" communication tone.
- **To integrate an automated de-identification and data privacy layer** that masks Personally Identifiable Information (PII) before processing queries, ensuring compliance with clinical data protection standards.
- **To establish a Role-Based Access Control (RBAC) mechanism** within the chatbot interface, restricting data retrieval so that doctors can only access and query reports of patients specifically assigned to them.
- **To implement a comprehensive audit logging and encryption system** for all conversational exchanges, providing a transparent and secure record of clinical decision-support interactions.
- **To evaluate the system's efficiency** by measuring the time taken for a doctor to identify abnormal clinical values using the chatbot versus manual report review, aiming for a significant improvement in diagnostic speed.

3. METHODOLOGY

3.1 Research Approach

This research adopts the **Design Science Research (DSR)** approach. Since the goal is to create an innovative "artifact" (the Doctor-Oriented Chatbot) to solve a specific clinical problem, DSR allows for an iterative process of designing, developing, and evaluating the system. The research focuses on "Experimental Development," where the chatbot's accuracy and security are tested against traditional manual report-reading methods.

3.2 Technical Methods and Algorithms

This research leverages an **In-context Learning** approach via RAG, utilizing pre-trained transformer-based architectures for clinical reasoning without the need for additional parametric fine-tuning.

The core functionality of the chatbot is built on the following technical frameworks:

- **Retrieval-Augmented Generation (RAG):** This is the primary architecture. Instead of relying on the LLM's internal memory, the system retrieves relevant text chunks from the patient's verified medical reports using Vector Embeddings and feeds them as context to the generator.
- **Vector Similarity Search (Cosine Similarity):** Used to find the most relevant clinical information from the uploaded reports based on the doctor’s query.
- **Natural Language Processing (NLP) & Prompt Engineering:** Custom "System Prompts" are used to enforce a professional medical tone, handle clinical abbreviations (e.g., CBC, LFT), and ensure the bot acts as an assistant rather than a diagnostic tool.
- **De-identification Algorithm:** A rule-based/NER (Named Entity Recognition) layer to identify and mask PII (Names, NIC, Phone numbers) before the data reaches the LLM.

3.3 Tools and Platforms with Justification

Table 2. Tools and platforms

Tool / Platform	Justification
Python / LangChain	Industry-standard framework for orchestrating RAG pipelines and managing complex LLM workflows.
Azure OpenAI (GPT-4o)	Provides state-of-the-art clinical reasoning capabilities with enterprise-grade data residency and privacy compliance.

Supabase Vector (pgvector)	Store "embeddings" directly in your PostgreSQL database using the pgvector extension.
Supabase Storage	used for your PDFs, images, and reports. It includes built-in CDN and image transformation.
FastAPI	Serves as the high-performance backend framework to develop RESTful APIs, facilitating secure, asynchronous communication between the React-based dashboard and the Azure RAG pipeline.
React.js & CSS	For developing a modern, responsive dashboard designed for clinical efficiency, featuring dark mode and high-contrast UI elements.
Supabase Auth	Handles logins (Email, Google, GitHub, etc.). It integrates directly with database via Row Level Security (RLS).

3.3.1 Frontend Design and User Experience (UX) Strategies

Implementation Considerations for Clinical AI Interface

To ensure clinical efficiency and a high-quality user experience (UX), the integration between the React frontend and the RAG backend focuses on the following research-driven technical aspects:

- **Streaming Responses (Token-based Streaming)**: To mimic natural conversation and reduce perceived wait times, the interface implements Server-Sent Events (SSE). This allows the LLM to stream the response character-by-character (or token-by-token) as it is generated, rather than waiting for the entire processing cycle to complete. This is critical for maintaining doctor engagement during complex report syntheses.
- **Source Citation and Transparency (Explainable AI):** A key differentiator of this RAG-based system is Traceability. When the chatbot provides a summary, the interface dynamically displays Source Citations (e.g., a small PDF icon or a clickable link next to the statement). This allows the clinician to instantly verify the AI-generated insight against the specific page or section of the original laboratory report, ensuring data integrity.
- **Latency Management and Clinical Feedback:** Given that RAG pipelines involve multiple steps (Retrieval, Context Injection, and Generation), the interface utilizes

Asynchronous State Management. While the backend processes the query, the frontend displays a "Clinical Reasoning" or "Analyzing Reports..." loader. This provides immediate feedback to the doctor, preventing "System Freeze" misconceptions during high-latency operations.

3.4 Data Collection and Ethical Considerations

- **Data Source:** The system will be tested using **anonymized, synthetic medical datasets** (e.g., from Kaggle or MIMIC-III) and sample reports provided by collaborating clinical mentors to ensure no real patient data is at risk during the research phase.
- **Ethical Considerations: Informed Consent:** Access is strictly limited to verified doctors.
    - **Data Integrity:** The chatbot includes clear **disclaimers** stating it is a decision-support tool, not a diagnostic replacement.
    - **Privacy:** Implementation of **Role-Based Access Control (RBAC)** ensures doctors only see data for patients assigned to them.

3.5 Validation Strategy and Performance Metrics

The system will be validated through a comparative study involving medical students/interns:

- **Performance Metrics:**
    1. **Hallucination Rate:** Percentage of AI responses not supported by the source document (Target: <2%).
    2. **Response Latency:** The time taken to generate a summary (Target: <3 seconds).
    3. **Efficiency Gain:** Comparing the time a doctor takes to find an "abnormal value" manually vs. using the chatbot.
    4. **Clinical Accuracy:** Evaluated by a panel of medical professionals based on a Likert scale (1–5) for tone, terminology, and summary precision.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The **AI-Powered Medical Report Interpretation System** is designed as a modular, cloud-native ecosystem. While the system operates as a unified platform, the architecture is decoupled into four primary functional blocks to ensure security, data integrity, and specialized AI processing:

- **Identity & Access Layer (Member 1):** Managed via Superbase Auth to ensure only licensed medical professionals enter the system.
- **Clinical Data Management (Member 2):** A synchronized database (SQL/NoSQL) that stores patient profiles and maintains an audit trail of all record modifications.
- **AI-Assisted Analysis Engine (Member 4):** The "Parser" that extracts entities, numeric values, and lab reference ranges from unstructured PDFs.
- **Doctor-Oriented Clinical AI Chatbot (Member 3 - Current Focus):** The conversational intelligence layer that utilizes RAG to bridge the gap between extracted data and the doctor's specific clinical queries.

4.2 System Diagram of Components

1. **User Tier:** React.js Dashboard.
2. **API Tier:** FastAPI (The Hub).
3. **Logic Tier:** LangChain Orchestrator (RAG Pipeline) + Azure OpenAI.
4. **Data Tier:** Supabase Vector+ Superbase Storage + SQL (Audit Logs).

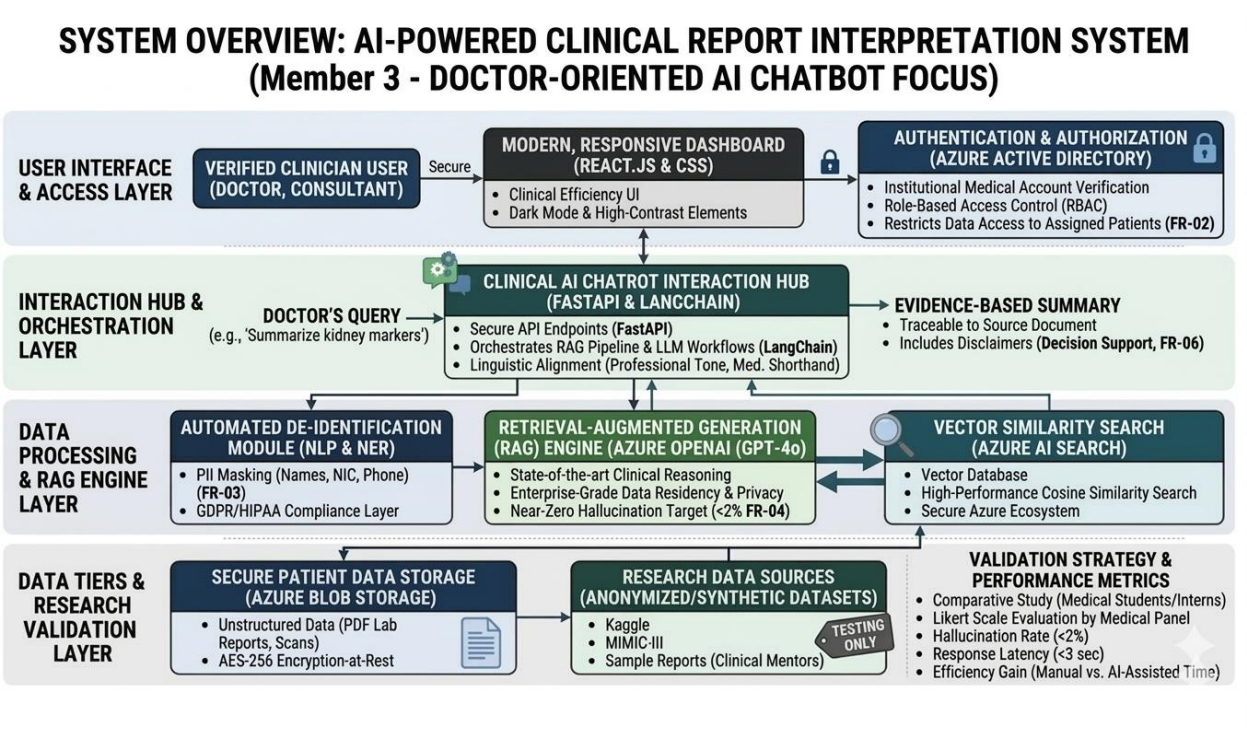


Figure 1. System Overview: AI-Powered Clinical Report Interpretation System

4.3 Individual Component and its Parts

The Clinical AI Chatbot is not a simple wrapper around an LLM; it is composed of several specialized sub-modules:

1. **Preprocessing & De-identification Unit:** Uses Named Entity Recognition (NER) to mask PII (Names, Phone numbers) before any data is sent to the LLM cloud, ensuring HIPAA/GDPR compliance [10].

2. **Vector Retrieval Unit:** Powered by AI Search, this part converts medical report text into mathematical vectors (embeddings) to find "contextually relevant" snippets based on a doctor's query [7].

3. **Prompt Engineering & Linguistic Alignment:** A "System Instruction" layer that forces the LLM to use professional medical shorthand (e.g., "CrCl", "BUN") and maintain a neutral, non-diagnostic tone [8].

4. **Reference Attribution Engine:** A verification step that maps every claim made by the chatbot back to a specific page or section of the uploaded medical report to eliminate hallucinations [13].

4.4 Component Interactions

The Chatbot (Member 3) acts as the primary "Interaction Hub." Its workflow depends on the other members:

- **Validation:** It queries Member 1 to verify the doctor's **RBAC** level before displaying any patient-specific insights [16].
- **Context Fetching:** It pulls the "Cleaned" clinical data from Member 4's analysis engine to populate the RAG context window.
- **Logging:** Every query and response is sent to Member 2's audit log to maintain clinical accountability [17].

4.5 Scalability and Security Considerations

- **Scalability:** By using **Azure OpenAI** and **FastAPI**, the system can handle concurrent queries from multiple departments without latency degradation. The use of vector indexing ensures that as the patient database grows, search retrieval remains or near-constant time.
- **Security:**
    - **Data-at-Rest:** All clinical reports in Blob Storage and vectors in AI Search are encrypted using AES-256.
    - **Zero-Retention Policy:** The system is configured so that clinical data sent to the LLM for transient processing is not used for training the global model, preserving institutional privacy [10].
    - **Auditability:** Every interaction is timestamped and linked to a Medical License ID, ensuring a "paper trail" for every clinical decision support action [17].

4.6 Data Flow Explanation

1. **Trigger:** A verified doctor enters a query (e.g., "Summarize any abnormal kidney function markers from the last 3 months").
2. **Security Check:** The system verifies the doctor is assigned to that specific patient via the RBAC module.
3. **Augmentation:** The query is converted into a vector. The system searches **Azure AI Search** for relevant chunks of that patient's reports.

4. **Generation:** The relevant clinical chunks + the doctor's query are sent to the **GPT-4o** model via an encrypted (TLS 1.3) tunnel.
5. **Output:** The chatbot generates a professional summary, highlighting the source of the data, and displays it on the React.js frontend.

Detailed Operational Workflow (Data Flow)

To illustrate the interaction between system components, consider the following clinical scenario: **Doctor's Query:** *"Is Patient A's blood pressure in the latest report lower or higher than the previous one?"*

The system executes the following sequential workflow:

1. **Intent Identification & Identity Check (Member 3 & 1):** The Chatbot (Member 3) receives the natural language query. It immediately communicates with the **Identity & Access Layer (Member 1)** to verify the logged-in doctor's credentials and their **Role-Based Access Control (RBAC)** permissions.
2. **Authorization Validation (Member 1):** The system checks the database to confirm if the doctor is officially assigned to **Patient A**.
    - **If Unauthorized:** The system denies the request and logs a security alert.
    - **If Authorized:** The process moves to the data retrieval phase.
3. **Context Retrieval (Member 3 & 4):** Once authorized, the Chatbot (Member 3) requests the processed clinical data from the **AI-Assisted Analysis Engine (Member 4)**. Member 4 provides the structured data points (current and historical Blood Pressure values) extracted from the patient's PDF reports stored in the secure cloud.
4. **Data Processing & De-identification:** The Chatbot engine applies the **PII Masking layer** to ensure that any personal identifiers are hidden before the clinical values are sent to the LLM (GPT-4o) for comparative analysis.
5. **Grounded Response Generation (RAG):** Using the **Retrieval-Augmented Generation (RAG)** framework, the Chatbot compares the latest value against the historical baseline provided by Member 4. It generates a professional, neutral summary (e.g., *"Patient A's systolic BP has decreased by 10mmHg compared to the report dated 20/01/2026"*).
6. **Audit Logging (Member 2):** Finally, the entire interaction—including the doctor's ID, the query, and the AI's response—is sent to the **Clinical Record Management system (Member 2)** to be stored in an encrypted audit log for clinical accountability.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The successful implementation and operation of the AI-Powered Medical Report Interpretation System involve several key stakeholders:

1. **Primary Stakeholders (Medical Doctors):** These are the core users of the system, including consultants, senior medical officers, and medical interns. Their primary need is to quickly synthesize complex lab data into actionable clinical insights.
2. **Secondary Stakeholders (Patients):** While patients do not interact with the system directly, they are the ultimate beneficiaries. Improved diagnostic speed and accuracy directly impact their safety and treatment outcomes.
3. **Tertiary Stakeholders (Hospital IT Administrators):** Responsible for managing the Azure environment, overseeing Member 1's doctor qualification workflow, and ensuring the technical health of the infrastructure.
4. **Regulatory Bodies:** Entities ensuring that the AI implementation complies with healthcare data standards (e.g., SLMC, Ministry of Health, and international standards like GDPR/HIPAA).

5.2 Functional Requirements (FR)

The following requirements define the specific behaviors and services the Member 3 (Chatbot) component must provide:

1. **FR-01: Secure Authentication Integration:** The system shall verify the doctor's identity via Azure Active Directory (Superbase Auth) before granting access to the chatbot interface.
2. **FR-02: RBAC-Enforced Data Retrieval:** The chatbot shall only retrieve and display clinical information for patients specifically assigned to the logged-in doctor [16].
3. **FR-03: Automated PII Masking:** The system shall automatically de-identify Personally Identifiable Information (PII) from the clinical text before it is processed by the AI engine [10].
4. **FR-04: Grounded Clinical Querying (RAG):** The chatbot shall generate responses strictly based on the retrieved context from uploaded medical reports, avoiding general-knowledge hallucinations [7].
5. **FR-05: Medical Shorthand Interpretation:** The NLP engine shall correctly interpret standard medical abbreviations (e.g., CBC, LFT, CrCl) and clinical units.

6. **FR-06: Evidence Attribution (Citations):** For every summary or insight provided, the chatbot shall cite the specific source document and page number used to generate the answer [13].
7. **FR-07: Interactive Trend Analysis:** The chatbot shall allow doctors to query trends across multiple historical reports (e.g., "Compare the last three creatinine levels").
8. **FR-08: Audit Trail Logging:** The system shall maintain a permanent, encrypted log of all doctor queries and AI responses for accountability and clinical review [17].

5.3 Non-Functional Requirements (NFR)

The following requirements define the quality attributes and constraints of the system:

- **Security & Privacy:**
    - **NFR-01:** All data in transit must be encrypted using TLS 1.3 or higher.
    - **NFR-02:** The system must comply with zero-retention policies where patient data is not used to train the underlying LLM [10].
- **Performance & Reliability:**
    - **NFR-03 (Latency):** The chatbot must generate a response to a standard clinical query within 3 seconds to ensure clinical efficiency.
    - **NFR-04 (Accuracy):** The system must maintain a hallucination rate of less than 2%, verified against source documents [13].
    - **NFR-05 (Availability):** The system shall maintain 99.9% uptime, utilizing Azure's cloud-native high-availability infrastructure.
- **Usability:**
    - **NFR-06 (Interface):** The dashboard must be optimized for high-pressure clinical environments, featuring a clean, distraction-free UI with dark mode support.
    - **NFR-07 (Responsiveness):** The interface must be fully responsive, allowing doctors to use the system on tablets or mobile devices during hospital rounds.
- **Scalability:**
    - **NFR-08:** The system must be capable of handling concurrent queries from multiple departments within a hospital without performance degradation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary market for this system is the **Digital Health and HealthTech sector**, specifically focusing on secondary and tertiary healthcare providers.

- **Primary Market:** Private and Government Hospital Chains in Sri Lanka and the South Asian region, where the patient-to-doctor ratio is high.
- **Customer Persona (The Decision Maker):** * **Medical Directors/CEOs:** Looking to reduce diagnostic errors and improve hospital efficiency.
    - **Chief Information Officers (CIOs):** Seeking secure, HIPAA-compliant AI solutions that integrate with existing Hospital Information Systems (HIS).
    - **Consultants & Private Practitioners:** Individual specialists who manage high volumes of lab reports and need a "digital co-pilot" for rapid analysis.

6.2 Value Proposition

Our system offers a unique value proposition centered on **"Professional-Grade Reliability"** rather than general AI assistance:

- **Zero-Hallucination Guarantee:** Unlike ChatGPT, our RAG-based approach ensures every word is backed by a verified clinical report.
- **Clinical Efficiency:** Reduces the time spent on manual data synthesis by over 60%, allowing doctors more time for patient care.
- **Security-First Compliance:** Built-in PII masking and Azure-grade encryption ensure patient data never leaves the secure institutional boundary.

6.3 Revenue Model

The project will follow a **B2B (Business-to-Business) SaaS model**:

- **Subscription Fee:** Tiered monthly or annual plans based on the number of active medical users (e.g., Small Clinics vs. Large Multi-specialty Hospitals).
- **Implementation & Integration Fee:** One-time cost for integrating the chatbot with the hospital's existing EHR/HIS databases.
- **Maintenance & Support:** Annual contracts for system updates, security patches, and AI model fine-tuning.

6.4 Cost Estimation (Initial Phase)

Table 3. cost estimation

Category	Estimated Cost (USD/Year)	Justification
Infrastructure	LKR 45,000 - 90,000	Azure OpenAI (GPT-4o), Azure AI Search, and Blob Storage.
Development	LKR 150,000 - 250,000	Engineering costs for RAG pipeline optimization and NLP fine-tuning.
Security Audits	LKR 30,000	Regular penetration testing and HIPAA/GDPR compliance checks.
Marketing	LKR 15,000	Presence at medical conferences and digital outreach to hospital boards.

6.5 Pricing Strategy

- **Freemium/Pilot Version:** A 3-month trial for a single department (e.g., Cardiology) to demonstrate efficiency gains.
- **Per-User Model:** Approximately LKR 6000 – LKR 10000 per doctor/month, making it affordable for both individual consultants and large institutions.
- **Enterprise Licensing:** Custom pricing for government-level deployments with unlimited patient records.

6.6 Competitive Advantage

Compared to general LLMs and existing medical scribes, our system holds the following edges:

1. **Medical Linguistic Precision:** Specifically tuned for shorthand (CBC, CrCl, LFT) which general bots often misinterpret.
2. **Traceability:** Every response includes a "source citation," making it a defensible tool in a clinical setting.
3. **Privacy-First Design:** Automated de-identification (PII masking) is a native feature, not an afterthought.

6.7 Intellectual Property (IP) Considerations

- **Proprietary Algorithms:** The specific "De-identification Layer" and "Prompt Engineering Framework" will be protected as trade secrets.
- **Software Copyright:** The full source code of the RAG orchestrator and the React dashboard will be copyrighted under the parent organization/team.
- **Trademark:** Registering a unique brand name (e.g., "MedSecure AI" or "ClinicalLink") to establish market identity.
- **Data Agreements:** Strict "Zero-Retention" contracts with LLM providers (Azure OpenAI) ensure that the institutional IP (patient data) is never used for external training.

7.BUDGET AND JUSTIFICATION

Table 4. budget and justification

Item	Estimated Cost (LKR)	Justification
LLM API Usage (OpenAI - GPT-4o)	LKR 6,000 - 8,000	Required for medical intent parsing, RAG-based response generation, and testing clinical summaries.
Superbase (Search & Storage)	LKR 0 (Open Source)	Hosting the Vector Database (Azure AI Search) and Blob Storage for patient report PDFs.
Software Licences	LKR 0 (Open Source)	Python, LangChain, FastAPI, and React.js are open-source frameworks.
Human Effort	LKR 0 (Academic)	Research and development effort contributed by the student as part of the degree requirement.
Miscellaneous	LKR 3,000 – 5,000	Documentation, report printing, and incidental testing costs.
TOTAL	**LKR 25,000 – 35,000**	

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Individual Task Breakdown

The technical scope for Member 3 focuses on the **Conversational Intelligence Layer**, ensuring the system provides secure, grounded, and clinically aligned responses.

- **T 3.1: RAG Pipeline Orchestration** – Design and implementation of the retrieval logic using LangChain and Azure OpenAI to fetch context from patient data.
- **T 3.2: PII De-identification Engine** – Development of an automated masking layer to redact Personally Identifiable Information before LLM processing.
- **T 3.3: Clinical NLP and Prompt Engineering** – Optimization of system instructions for Doctor-to-Doctor tone and medical abbreviation interpretation.
- **T 3.4: Evidence Attribution and Source Mapping** – Implementation of a citation engine to map AI summaries to specific source document pages.
- **T 3.5: Clinical Dashboard Interface** – Development of the user-facing chat interface using React.js and Tailwind CSS.

8.2 Timeline and Task Alignment

Table 5. Timeline and task alignment

Week	Phase	Task ID	Activity Description
01 - 02	Research & Design	**T 3.1**	Architectural mapping of the RAG pipeline and retrieval logic requirements.
03 - 04	Security Integration	**T 3.2**	Deployment of the PII masking and data de-identification algorithms.
05 - 08	AI Development	**T 3.1, T 3.4**	Construction of the RAG engine and implementation of the source mapping/citation system.
09 - 10	NLP Refinement	**T 3.3**	Fine-tuning the NLP engine for medical linguistic alignment

			and shorthand handling.
11	System Integration	**T 3.5**	Development of the React.js dashboard and establishing API connectivity with other modules.
12	Validation	All Tasks	Final performance evaluation, measuring latency and hallucination rates for reporting.

8.3 Workload Distribution Logic

The workload is distributed to ensure that Member 3 serves as the Interaction Hub of the system. While foundational infrastructure is handled by other members, Member 3 manages the critical logic governing the safety, accuracy, and professional presentation of clinical insights. This modular approach ensures that the intelligent response system remains decoupled from the data extraction process, allowing for specialized optimization of the AI's reasoning capabilities.

9.GANTT CHART

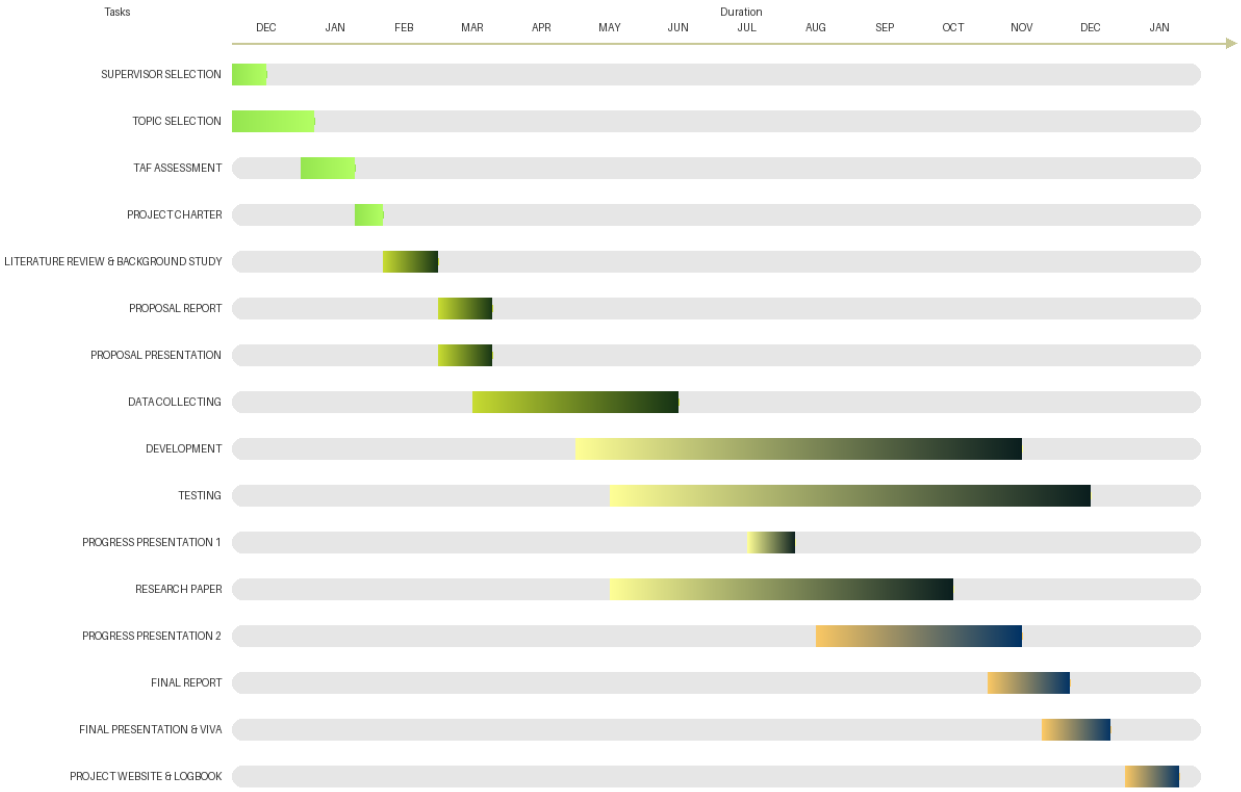


Figure 2. Gantt Chart

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